

LAB NO : 02

LAB TITLE : Basic Syntax and Input/Output in C++

OBJECTIVE :

The objective of this lab is to understand the basic structure of a C++ program. In this lab we learn how to use input and output statements, work with different data types, apply indentation, and perform calculations using arithmetic operators.

THEORY :

C++ programs follow a specific structure that includes header files, a main function, and program statements. The iostream library is used for input and output operations. The cout statement is used to display output on the screen, while cin is used to take input from the user. The symbols << and >> are known as stream insertion and extraction operators. C++ is a statically typed language, which means every variable must be declared with a data type before it is used. Some common data types include: • int for integers • double for decimal numbers • char for single characters • bool for true/false values • string for text C++ also provides arithmetic operators such as +, -, *, /, and % which are used to perform mathematical calculations. Proper indentation makes the program easier to read and understand. Comments are used to explain the code and can be written using // for single line comments or /* */ for multi-line comments.

SOFTWARE :

1. DEV-C++ IDE
2. C++ Compiler

PROCEDURE / PROCESS :

Start the program Hello World and User Greeting :

1. Start the program Hello World and User Greeting

2. Declare a variable to store the user's name
3. Ask the user to enter their name
4. Display a greeting message
5. End the program

```
#include <iostream> // input/output ke liye library
using namespace std; // standard namespace use karne ke liye

int main ()
{
    string name; // string variable jisme user ka naam store hoga

    cout << "Enter your name: "; // user se naam input lene ka message
    cin >> name; // user ka naam input lena

    cout << "Hello " << name << "! Welcome to C++ programming."; // user ko
    welcome message display karna

    return 0; // program successful execution ke baad end ho raha hai
}
```

Simple Calculator Using Arithmetic Operators :

1. Start the program.
2. Declare two numbers.
3. Take input from the user.
4. Perform addition, subtraction, multiplication, and division.
5. Display the results.
6. End the program.

```
#include <iostream> // input aur output ke liye library
using namespace std; // standard namespace use karne ke liye

int main()
{
```

```
int a, b; // do integer variables declare kiye gaye hain

cout << "Enter two numbers: "; // user se do numbers input lene ka message
cin >> a >> b; // user se values lena

cout << "Addition = " << a + b << endl; // addition ka result display
karna
cout << "Subtraction = " << a - b << endl; // subtraction ka result
display karna
cout << "Multiplication = " << a * b << endl; // multiplication ka result
display karna
cout << "Division = " << a / b << endl; // division ka result display
karna

return 0; // program successful execution ke baad end ho raha hai
}
```

CONCLUSION

In this lab we learned the basic structure of a C++ program. We also learned how to use input and output statements, work with different data types, and perform calculations using arithmetic operators. This lab helped us understand the fundamentals of C++ programming.